



# **Signals and Systems**

## **Discrete Time Fourier Series**

**(lecture 24, Date:-06/04/2026)**

**By: Dr. Sajai Vir Singh**

# Fourier Transform of Discrete Time Aperiodic signal (DTFT)

❖ Transform the discrete time signal  $x(n)$  from time domain into frequency domain  $X(\omega)$  is called DTFT.

$$F[x(n)] = X(e^{j\omega}) = X(\omega)$$

$$F^{-1}[X(\omega)] = x(n)$$

$$x(n) \xleftrightarrow{F} X(\omega)$$

**DT Fourier Transform Pair**

$$F[x(n)] = X(e^{j\omega}) = X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

(analysis equation)

$$x[n] = F^{-1}[X(\omega)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

(synthesis equation)

OR

$$x[n] = F^{-1}[X(\omega)] = \frac{1}{2\pi} \int_0^{2\pi} X(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

- ❖ For the continuous time signals, the Fourier transform and the spectrum of the signal have a frequency range of  $(-\infty, +\infty)$
- ❖ while for the discrete time signals, the Fourier transform and the spectrum of the signal have a frequency range of  $(-\pi, +\pi)$  or  $(0, 2\pi)$

**Note:-** Fourier transform of discrete time signal is periodic with period  $2\pi$  while Fourier transform of continuous time signal is aperiodic .

**Proof:-**

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$$\begin{aligned} X(\omega + 2\pi) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega+2\pi)n} = \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega)n}e^{-j(2\pi)n} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega)n} = X(\omega) \end{aligned}$$

❖ hence, the Fourier transform  $X(\omega)$  is periodic with period  $2\pi$ .

## *Convergence*

The equation  $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$  converges either if  $x[n]$  is absolutely summable, that is

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty ,$$

or if the sequence has finite energy, that is

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty .$$

And there is no convergence issues associated with the synthesis equation

## Unit Impulse Functions

Example 1: Unit Impulse Functions  $x[n] = \delta[n]$

$$F[x(n)] = X(\omega) = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n} = 1$$

Example 2: Time shifted form of Unit Impulse Functions

$$x[n] = \delta[n - n_0]$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n - n_0]e^{-j\omega n} = e^{-j\omega n_0}$$

$$\left| X(e^{j\omega}) \right| = \left| e^{-j\omega n_0} \right| = 1 \quad \text{and} \quad \angle X(e^{j\omega}) = \angle e^{-j\omega n_0} = -\omega n_0$$

Amplitude spectrum

Phase spectrum

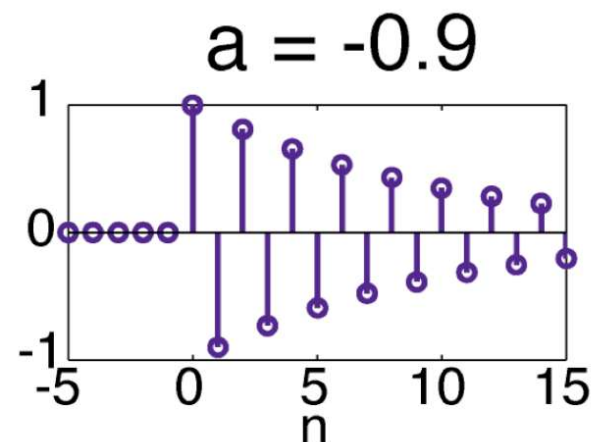
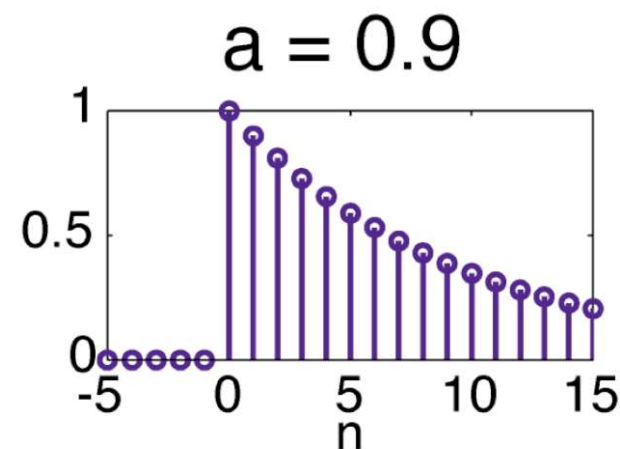
**Example 3:**  $x[n] = \delta[n] + \delta[n-2] + \delta[n+3] = (1, 0, 0, 1, 0, 1)$

$$\begin{aligned} X(\omega) &= F[x(n)] = F[\delta(n) + \delta(n-2) + \delta(n+3)] \\ &= F[\delta(n)] + F[\delta(n-2)] + F[\delta(n+3)] \\ &= 1 + e^{-j2\omega} + e^{j3\omega} \end{aligned}$$

**Example 4: Exponential Decay**

$$x[n] = a^n u[n] \quad |a| < 1$$

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}} \end{aligned}$$



## Amplitude Spectrum :

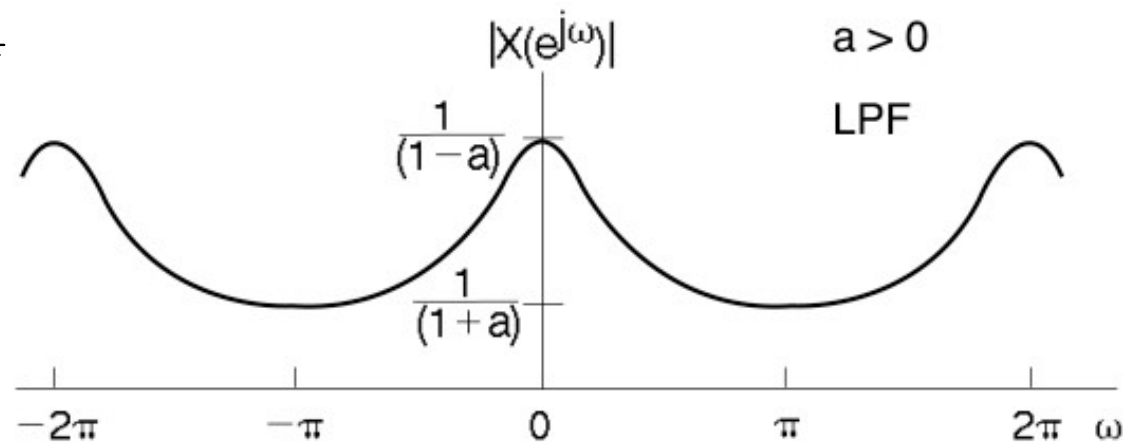
$$\begin{aligned}|X(\omega)| &= \left| \frac{1}{1 - ae^{-j\omega}} \right| = \left| \frac{1}{(1 - a \cos(\omega)) - ja \sin(\omega)} \right| \\&= \frac{1}{\sqrt{(1 - a \cos(\omega))^2 + (a \sin(\omega))^2}} \\&= \frac{1}{\sqrt{1 - 2a \cos(\omega) + a^2 \cos^2(\omega) + a^2 \sin^2(\omega)}} \\&= \frac{1}{\sqrt{1 - 2a \cos(\omega) + a^2}}\end{aligned}$$

For a=1:

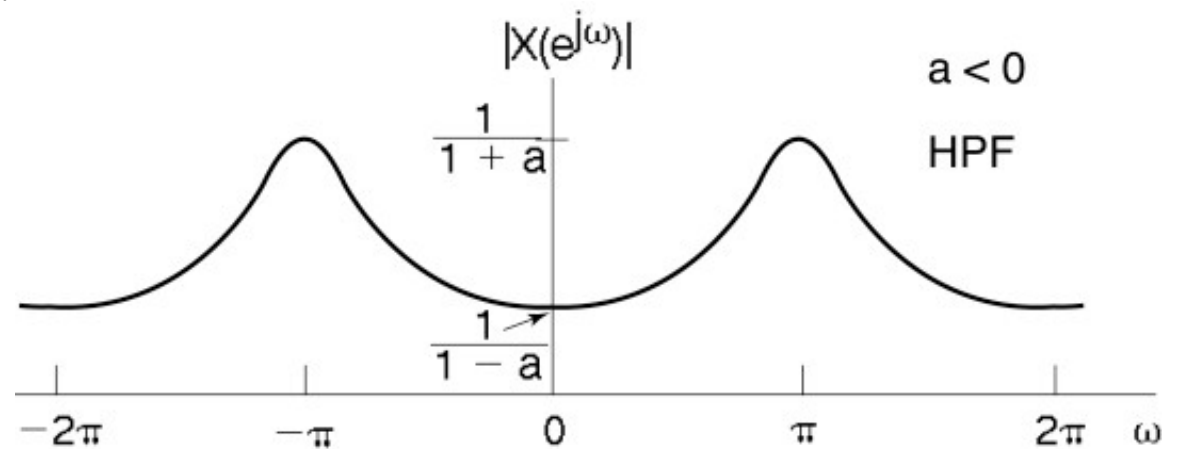
$$\begin{aligned}X(\omega) &= \frac{1}{\sqrt{1 - 2a \cos(\omega) + a^2}} = \frac{1}{\sqrt{2 - 2 \cos(\omega)}} = \frac{1}{\sqrt{2} \sqrt{1 - (1 - 2 \sin^2(\omega / 2))}} \\&= \frac{1}{2 \sin(\omega / 2)}\end{aligned}$$



$$\begin{aligned}
 \left|X(e^{j\omega})\right|_{\omega=0} &= \frac{1}{\sqrt{1-2a\cos(0)+a^2}} \\
 &= \frac{1}{\sqrt{(1-2a+a^2)}} \\
 &= \frac{1}{\sqrt{(1-a)^2}} \\
 &= \frac{1}{1-a}
 \end{aligned}$$



$$\begin{aligned}
 \left|X(e^{j\omega})\right|_{\omega=\pi} &= \frac{1}{\sqrt{1-2a\cos(\pi)+a^2}} \\
 &= \frac{1}{\sqrt{(1+2a+a^2)}} \\
 &= \frac{1}{\sqrt{(1+a)^2}} \\
 &= \frac{1}{1+a}
 \end{aligned}$$



Phase Spectrum :

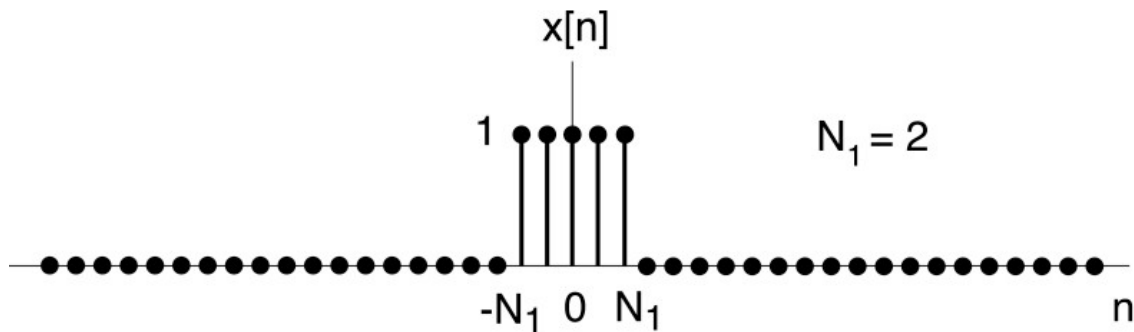
$$\angle X(\omega) = \tan^{-1}\left(\frac{a\sin(\omega)}{1 - a\cos(\omega)}\right)$$

For a=1:

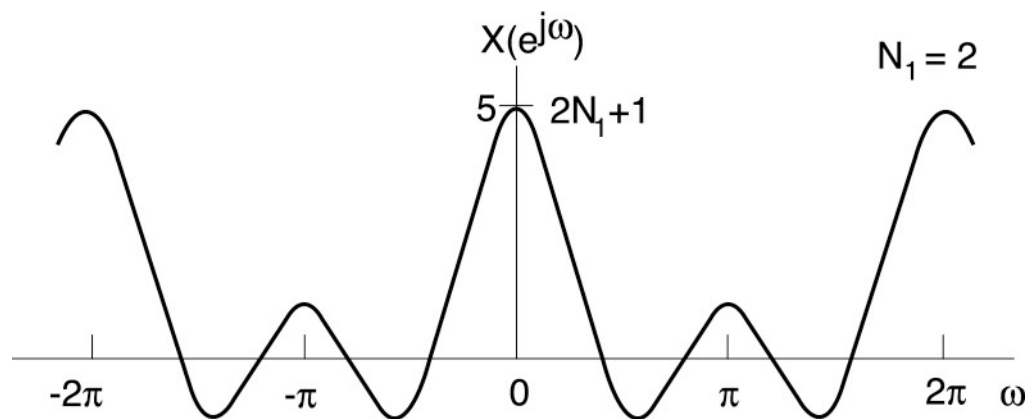
$$\begin{aligned}\angle X(\omega) &= \tan^{-1}\left(\frac{\sin(\omega)}{1 - \cos(\omega)}\right) = \tan^{-1}\left(\frac{2\sin(\omega/2)\cos(\omega/2)}{2\sin^2(\omega/2)}\right) \\ &= \tan^{-1}\left(\frac{\cos(\omega/2)}{\sin(\omega/2)}\right) = \tan^{-1}(\cot(\omega/2)) \\ &= \tan^{-1}(\tan(\pi/2 - \omega/2)) = (\pi/2 - \omega/2)\end{aligned}$$

Example 5:

$$x[n] = \begin{cases} 0, & n < -N_1 \\ 1, & -N_1 \leq n \leq N_1 \\ 0, & n > N_1 \end{cases}$$



$$X(\omega) = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \sum_{n=-N_1}^{N_1} (e^{-j\omega})^n = \frac{\sin(\omega(N_1 + 1/2))}{\sin(\omega/2)}$$



Example 6:

$$x[n] = a^{|n|}, \quad |a| < 1$$

Solution:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^{|n|} e^{-j\omega n} = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

Let  $m = -n$  in the first summation, we obtain

$$\begin{aligned} X(e^{j\omega}) &= \sum_{m=1}^{\infty} a^m e^{j\omega m} + \sum_{n=0}^{\infty} a^n e^{-j\omega n} \\ &= \frac{ae^{j\omega}}{1 - ae^{j\omega}} + \frac{1}{1 - ae^{-j\omega}} \\ &= \frac{1 - a^2}{1 - 2a \cos \omega + a^2} \end{aligned}$$

**Example 7:**  $x[n] = 3^n u[n]$       **OR**       $x[n] = a^n u[n] \quad |a| > 1$

The given sequence is not absolutely summable. Therefore, its DTFT does not exist.

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{\infty} 3^n = 1 + 3 + 3^2 + 3^3 + 3^4 + 3^5 + \dots = \text{infinite}$$

**Example 8:**  $x[n] = (0.5)^n u[n] + 2^n u(-n-1)$

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} [(0.5)^n u[n] + 2^n u(-n-1)] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (0.5)^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} 2^n e^{-j\omega n} = \sum_{n=0}^{\infty} (0.5 e^{-j\omega})^n + \sum_{n=1}^{\infty} (2^{-1} e^{j\omega})^n \\ &= \frac{1}{1 - 0.5 e^{-j\omega}} + \frac{2^{-1} e^{j\omega}}{1 - 2^{-1} e^{j\omega}} = \frac{1}{1 - 0.5 e^{-j\omega}} - \frac{1}{1 - 2 e^{-j\omega}} \end{aligned}$$

Thank You